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Computational Fluid Dynamics Analysis on NACA 0024 (naca0024-il)

Abstract

This report will use a Computational Fluid Dynamics (CFD) simulation technique to analyse the aerodynamic performance of NACA 0024 (naca0024-il). The NACA 0024 is a symmetrical aerofoil with a maximum thickness of 24% at a 30% chord *(Refer to Reference 1)*. I will be using a programme called Ansys to produce an aerodynamic flow data on the different axis.
I will be completing a mesh and domain study to determine the best parameters needed for more exact result. Using the best parameters, I will then simulate the aerofoil at multiple angles of attack at a subsonic free stream velocity. This will also be done for a transonic and supersonic free stream velocity.

Set-up, Methodology

I will study and discuss the aerodynamic performance of a 2D and 3D geometry of NACA 0024 aerofoil. Before starting the analysis I needed to find experimental data *(Refer to Reference 2*) on the aerofoil to compare my results. I used the experimental data to set up the CFD so that I was able to get similar results. When preparing for the CFD I set the chord to 1800mm, the density at 1.204 Kg/m^3 and a span of 30mm. In Task 1 and 2 I set the aerofoil 0 degrees. The subsonic free stream velocity was set to 50m/s (which Mach = 0.150602) with a Reynolds number of 6.00e+6. The transonic free stream velocity was set at 250m/s (which Mach = 0.753012) and the supersonic free stream velocity at 450m/s (which Mach = 1.355422). the formular for Mach number is:

2D Aerofoil	3D Aerofoil/Wing
$Cl = \frac{L}{(0.5 \times \rho \times V^2 \times C \times distance\ between\ the\ two\ symmetry\ faces)}$	$Cl = \frac{L}{(0.5 \times \rho \times V^2 \times A)}$
$Cd = \frac{D}{(0.5 \times \rho \times V^2 \times C \times distance\ between\ the\ two\ symmetry\ faces)}$	$Cd = \frac{D}{(0.5 \times \rho \times V^2 \times A)}$
$Cm = \frac{M}{(0.5 \times \rho \times V^2 \times C^2 \times distance\ between\ the\ two\ symmetry\ faces)}$	$Cm = \frac{M}{(0.5 \times \rho \times V^2 \times C^2 \times S)}$
Where distance between the two symmetry faces is 30mm = 0.03m	Where S= 200mm or 2m

Table 1

Figure 1

$$Mach\ No. = \frac{U_\infty}{V_{Sound}}$$

Where U_∞ is Free stream velocity and V_{Sound} is Speed of sound. Then for task 4 I increase the span to 2000mm to mimic a 3D aerofoil whilst maintaining a subsonic free stream velocity of 50m/s.
For each of these tasks I will take the results a force at X, Y, and Z axes and torque on the Z axis. From these results I will use the formulas in figure 1 to calculate the coefficient of lift, the coefficient of drag and the coefficient of moment. These coefficients tell me how aerodynamic the aerofoil is.

Mesh Study (2D) -Task 1

Mesh 1		Mesh 2		Mesh 3		Mesh 4	
Inflation		**Inflation**		**Inflation**		**Inflation**	
First layer hieght	8.10E-06	First layer hieght	8.10E-06	First layer hieght	8.10E-06	First layer hieght	8.10E-06
Maximum Angle	180	Maximum Angle	180	Maximum Angle	180	Maximum Angle	180
Maximum layers	36	Maximum layers	23	Maximum layers	10	Maximum layers	15
Element size	0.03	Element size	0.025	Element size	0.06	Element size	0.08
Growth Rate	1.2	Growth Rate	1.2	Growth Rate	1.2	Growth Rate	1.2
Statistics		**Statistics**		Statistics		**Statistics**	
Nodes	69796	Nodes	126480	Nodes	19773	Nodes	12150
Elements	256279	Elements	472090	Elements	55878	Elements	33864
Quality (Average)		**Quality (Average)**		**Quality (Average)**		**Quality (Average)**	
Skewness	0.32692	Skewness	0.31747	Skewness	0.45186	Skewness	0.58212
Orthagonal Quality	0.67121	Orthagonal Quality	0.68134	Orthagonal Quality	0.54599	Orthagonal Quality	0.41384

Table 1

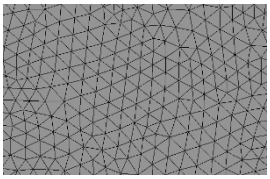


Figure 4

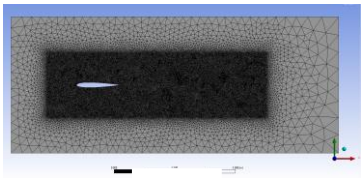


Figure 2

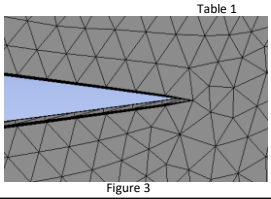


Figure 3

For the mesh study I decided to compare 4 different meshes. In each mesh there were inflation parameters such as the maximum layers, the element size and the growth rate can be adjusted. I initially used a website *(Refer to Reference 3)* need to find the first layer height. The online calculator required the free stream velocity, the density and the chord length to calculate y+. When adjusting the element size, I found that increasing it decreased the number of elements produced. When I first started creating the meshes, I changed the Growth rate to above 1.3. This had an impact on my results as the number of cells were very small which overall meant the mesh was too small. Thus, producing inaccurate results. This forced me to retrace my steps and use the recommended growth rate of 1.2. I found when using this growth rate running the solutions took longer as it had more cells to compute.
As shown in table 2 the discussed parameters have been adjusted. Figures 2, 3 and 4 are images of mesh 2. The inner domain is denser with cells to gauge more meticulous forces. In table 2 the statistics tells us the larger number of nodes and elements each mesh have, this helps me determine whether the mesh will give a more precise result or not. Figure 4 tells us whether the quality of the mesh is excellent or bad. I also found that if the Orthogonal quality is close to 1 them the quality of the mesh is very good.

Once I had ran the 4 different meshes through the CFX-solutions I was able to collect the amount of force that was applied to the aerofoil at the X, Y and Z axes. I used these values to calculate the lift, drag, coefficient of lift and the coefficient of drag. Refer to the experimental care and set up section for all equations used.
Overall, I found that Mesh 2 was the best mesh as it had the largest number elements, the skewness was rated good according to figure 5 and the orthogonal quality was roughly close to 1.

Value of Skewness	Cell Quality
1	degenerate
0.9 — <1	bad (sliver)
0.75 — 0.9	poor
0.5 — 0.75	fair
0.25 — 0.5	good
>0 — 0.25	excellent
0	equilateral

Skewness and the mesh quality table.

Figure 5

Results					
X(A) - Force (N)	0.729133	X(A) - Force (N)	0.70795	X(A) - Force (N)	1.32648	X(A) - Force (N)	1.209
Y(N) - Force (N)	-0.090479	Y(N) - Force (N)	-0.0271244	Y(N) - Force (N)	0.148277	Y(N) - Force (N)	-0.0778049
Z - Force (N)	-0.00310908	Z - Force (N)	-0.00344681	Z - Force (N)	-0.00773908	Z - Force (N)	-0.00550275
Z - Tourqe (Piching Moment) (Nm)	-0.0879424	Z - Tourqe (Piching Moment) (Nm)	-0.0342273	Z - Tourqe (Piching Moment) (Nm)	0.200677	Z - Tourqe (Piching Moment) (Nm)	-0.0730911
Area	0.018	Area	0.018	Area	0.018	Area	0.018
Lift	-0.090479	Lift	-0.0271244	Lift	0.148277	Lift	-0.0778049
Drag	0.729133	Drag	0.70795	Drag	1.32648	Drag	1.209
Cl	-0.00339109	Cl	-0.00100102	Cl	0.00547232	Cl	-0.002871374
Experimental Cl	0.05	Experimental Cl	0.05	Experimental Cl	0.05	Experimental Cl	0.05
Cl Percentage Difference %	106.6782179	Cl Percentage Difference %	102.0020408	Cl Percentage Difference %	89.05573547	Cl Percentage Difference %	105.7427477
Cd	0.026908504	Cd	0.02612675	Cd	0.048953472	Cd	0.044617897
Experimental Cd	0.0052	Experimental Cd	0.0052	Experimental Cd	0.0052	Experimental Cd	0.0052
Cd Percentage Difference %	-417.4712267	Cd Percentage Difference %	-402.4374907	Cd Percentage Difference %	-841.4129285	Cd Percentage Difference %	-758.0364804
Cm	-6.01018E-05	Cm	-2.33917E-05	Cm	0.000137147	Cm	-4.99521E-05

Table 2

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Domain Study (2D) – Task 2

Domain Size																	
Domain 1				Domain 2				Domain 3				Domain 4					
Outside Domain Size		Inside Domain Size		Outside Domain Size		Inside Domain Size		Outside Domain Size		Inside Domain Size		Outside Domain Size		Inside Domain Size			
Sides	Size (mm)	Sides	Size (mm)	Sides	Size (mm)	Sides	Size (mm)	Sides	Size (mm)	Sides	Size (mm)	Sides	Size (mm)	Sides	Size (mm)		
Font	2700	Font	1200	Font	1500	Font	750	Font	4000	Font	1800	Font	5200	Font	2600		
Top	2700	Top	1200	Top	1500	Top	750	Top	4000	Top	1800	Top	5200	Top	2600		
Bottom	2700	Bottom	1200	Bottom	1500	Bottom	750	Bottom	4000	Bottom	1800	Bottom	5200	Bottom	2600		
Back	9000	Back	6000	Back	3600	Back	1800	Back	13000	Back	7500	Back	18000	Back	9000		
Nodes		126480		Nodes		51446		Nodes		19773		Nodes		12150			
Elements		472090		Elements		168837		Elements		55878		Elements		33864			

Table 3

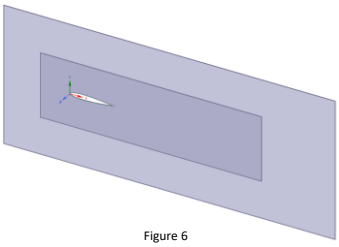


Figure 6

In the Domain study I used the best mesh (which was mesh 2) but changed the sizes of my domains in the geometry. I decided to analyse 4 different domain sizes. As shown in table 3 I have adjusted the outer and inner domain. It shows us that when increasing the domain sizes decreases the number of elements that are produced thus notifying me that the results that are generated will not be very accurate or like the experimental data from the study. When making the domain size smaller I found that the elements produced still decrease but not as significantly as when the domain sizes where increased. Therefore, the original domain size that I used will still make the most accurate results.

Results							
X(A) - Force (N)	0.70795	X(A) - Force (N)	0.835806	X(A) - Force (N)	0.788512	X(A) - Force (N)	1.93527
Y(N) - Force (N)	-0.0271244	Y(N) - Force (N)	-0.185694	Y(N) - Force (N)	0.0657546	Y(N) - Force (N)	0.115815
Z - Force (N)	-0.00344681	Z - Force (N)	0.00124433	Z - Force (N)	0.000606447	Z - Force (N)	0.000617055
Z - Tourqe (Piching Moment) (Nm)	-0.0342273	Z - Tourqe (Piching Moment) (Nm)	-0.2276	Z - Tourqe (Piching Moment) (Nm)	0.0695771	Z - Tourqe (Piching Moment) (Nm)	-0.063794
Area	0.018	Area	0.018	Area	0.018	Area	0.018
Lift	-0.0271244	Lift	-0.185694	Lift	0.0657546	Lift	0.115815
Drag	0.70795	Drag	0.835806	Drag	0.788512	Drag	1.93527
Cl	-0.00100102	Cl	-0.0068529999	Cl	0.0024266	Cl	0.004274129
Experimental Cl	0.05	Experimental Cl	0.05	Experimental Cl	0.05	Experimental Cl	0.05
Cl Percentage Difference %	102.0020408	Cl Percentage Difference %	113.705998	Cl Percentage Difference %	95.14667995	Cl Percentage Difference %	91.45174237
Cd	0.02612675	Cd	0.030845249	Cd	0.029099874	Cd	0.071420742
Experimental Cd	0.0052	Experimental Cd	0.0052	Experimental Cd	0.0052	Experimental Cd	0.0052
Cd Percentage Difference %	-402.4374907	Cd Percentage Difference %	-491.1778648	Cd Percentage Difference %	-459.6129539	Cd Percentage Difference %	-1273.475814
Cm	-2.33917E-05	Cm	-0.000155547	Cm	4.75505E-05	Cm	-4.35982E-05

Table 4

In table 4 the results for the different domain sizes are provided. From these results I can tell that domain 4 has the smallest Cl percentage difference which means that it has the closest results to the experimental data. Even though domain 4 is the closest to the experimental data we see that in table 3 the elements computed are very low so when changing the angle of attack there is a possibility the results maybe more untrue.

2D Results and their interpretations – Task 3

Angle on Attack - Subsonic

AoA							
0 Deg		2 Deg		6 Deg		10 Deg	
Result							
X(A) - Force (N)	0.729133	X(A) - Force (N)	1.35349	X(A) - Force (N)	2.44134	X(A) - Force (N)	12.7859
Y(N) - Force (N)	-0.090479	Y(N) - Force (N)	12.0706	Y(N) - Force (N)	34.9639	Y(N) - Force (N)	114.998
Z - Force (N)	-0.00310908	Z - Force (N)	-0.0014576	Z - Force (N)	-0.00123692	Z - Force (N)	0.0381926
Z - Tourqe (Piching Moment) (Nm)	-0.0879424	Z - Tourqe (Piching Moment) (Nm)	5.16914	Z - Tourqe (Piching Moment) (Nm)	15.1393	Z - Tourqe (Piching Moment) (Nm)	48.6178
Area	0.018	Area (M^2)	0.018	Area	0.018	Area	0.018
Lift	-0.090479	Lift	12.0706	Lift	34.9639	Lift	114.998
Drag	0.729133	Drag	1.35349	Drag	2.44134	Drag	12.7859
Cl	-0.003339109	Cl	0.445463017	Cl	1.290335557	Cl	4.243977599
Experimental Cl	0.05	Experimental Cl	0.18	Experimental Cl	0.5	Experimental Cl	0.85
Cl Percentage Difference %	106.6782179	Cl Percentage Difference %	-147.4794538	Cl Percentage Difference %	-158.0671114	Cl Percentage Difference %	-399.2914822
Cd	0.026908504	Cd	0.049950271	Cd	0.090097152	Cd	0.471861017
Experimental Cd	0.0052	Experimental Cd	0.0056	Experimental Cd	0.0075	Experimental Cd	0.012
Cd Percentage Difference %	-417.4712267	Cd Percentage Difference %	-791.9691212	Cd Percentage Difference %	-1101.295358	Cd Percentage Difference %	-3832.175138
Cm	-6.01018E-05	Cm	0.003532705	Cm	0.010346533	Cm	0.03322648

Table 5

After completing the mesh and domain study I used the best parameters from each to then produce results at different angles of attack whilst still at a subsonic free stream velocity of 50m/s. In this task we want to explore the way the Cl, Cd and Cm is affected. Table 5 is the results that were produced at an Angle of attack of 0, 2, 6 and 10 degrees. From these results I saw that the increase in angle of attack produced a higher Cl which became less similar to the results in the experimental data. This could have happened due to CFD not always being as accurate as a real-life scenario. Out of all the angles of attack the Cl percentage difference is lowest at 0 degrees meaning that it's the most related to the experimental data. Whereas the Cl percentage difference at 10 degrees is significantly higher. Therefore, it could imply that symmetrical wings like the NACA0024 are more likely to stall at higher angles of attack.

In the CFX results section I used the tools provided to create velocity vector as shown in Figure 7. This velocity vector represents what would happens to velocity around the aerofoil at an angle of attack of 6 degrees. In this figure I can see that above the aerofoil the vector lines change colour to red, this indicates that the velocity is increasing. Whereas behind the trailing edge the colour of the vector becomes greener which means that the velocity decrease.

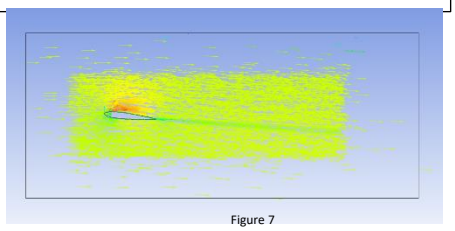


Figure 7

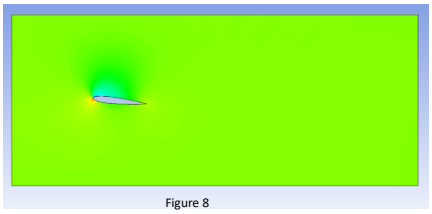


Figure 8

Figure 8 shows the pressure contours that occur on the aerofoil at an angle of attack of 6 degrees. It's clearly demonstrated that there is a high amount of pressure just underneath the front of the aerofoil. this high amount of pressure will increase the lift. There is a low amount pf pressure generated on the top of the aerofoil.

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Angle on Attack – Transonic

M-Transonic					
0 Deg		5 Deg		10 Deg	
X(A) - Force (N)	2.84E+01	X(A) - Force (N)	1.27E+02	X(A) - Force (N)	5.23E+02
Y(N) - Force (N)	-1.07E+00	Y(N) - Force (N)	1.46E+03	Y(N) - Force (N)	1.88E+03
Z - Force (N)	1.21E-02	Z - Force (N)	0.0793649	Z - Force (N)	0.00527969
Z - Tourqe (Piching Moment) (Nm)	-4.12E+00	Z - Tourqe (Piching Moment) (Nm)	5.37E+02	Z - Tourqe (Piching Moment) (Nm)	7.62E+02
Area	0.018	Area	0.018	Area	0.018
Lift	-1.07407	Lift	1456.41	Lift	1875.47
Drag	28.3587	Drag	127.261	Drag	522.527
Cl	-0.039638333	Cl	53.74851228	Cl	69.21383561
Cd	1.046572006	Cd	4.696541098	Cd	19.28375174
Cm	-2.81E-03	Cm	3.67E-01	Cm	5.21E-01

Table 6

Once testing different angles of attack at s subsonic free stream velocity I decided to compare how the coefficients of lift drag and moment were affected at a transonic free stream velocity. I settled on a velocity of 250m/s which is equal to a Mach number of 0.753012. at a higher velocity I expected to see a larger Cl compared to the subsonic velocity. This is clearly displayed in table 6, we can see that Cl has a significant increase between 0 degrees and 5 degrees whereas there is less of increase between 5 degrees and 10 degrees. Thus, potentially indicating the aerofoil could be close to stalling. We can all see that the drag to lift ratio is a lot small compared to the drag at subsonic velocity. Therefore, this indicates that the aerofoil has a greater lift when taking of at the transonic velocity. I also found that when running the solutions in CFX it would always fail, to fix this I adjusted the maximum layers in the mesh from 23 to 8. This fixed it.

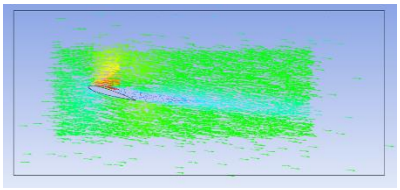


Figure 9

Similarly with the subsonic velocity I created a velocity vector for the transonic velocity at an angle of attack of 10 degrees as shown in figure 9. There is in increase in velocity above the wing which also occurs at subsonic velocity. although the velocity seems to be larger higher above the aerofoil compared to the subsonic velocity vector. There is also a gap behind the aerofoil where the vectors curl. This is also known as flow separation which decreases the amount of lift produced.

Figure 10 shows the pressure contours at an angle of attack at 10 degrees. Due to the velocity being higher the amount of pressure applied on the aerofoil is larger than the subsonic velocity. The pressure applied on the top is larger. But it is less distributed along the top.

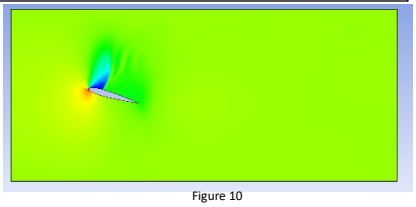


Figure 10

Angle on Attack – Supersonic

M - Supersonic					
0 Deg		5 Deg		10 Deg	
X(A) - Force (N)	824.866	X(A) - Force (N)	1393.01	X(A) - Force (N)	2617.37
Y(N) - Force (N)	7.9926	Y(N) - Force (N)	3025.83	Y(N) - Force (N)	6869.06
Z - Force (N)	0.0868565	Z - Force (N)	0.0137674	Z - Force (N)	0.00586906
Z - Tourqe (Piching Moment) (Nm)	15.7813	Z - Tourqe (Piching Moment) (Nm)	2395.66	Z - Tourqe (Piching Moment) (Nm)	5621.23
Area	0.018	Area	0.018	Area	0.018
Lift	7.9926	Lift	3025.83	Lift	6869.06
Drag	824.866	Drag	1393.01	Drag	2617.37
Cl	0.294965263	Cl	111.6676354	Cl	253.5012501
Cd	30.44151051	Cd	51.40874828	Cd	96.59350291
Cm	1.08E-02	Cm	1.64E+00	Cm	3.84E+00

Table 7

For the supersonic velocity I found that I had the same issue as transonic, where the solutions in CFX would not work. After adjusting the maximum layers, I was able to run 100 iterations without failure. I set my supersonic velocity to 450m/s (Mach = 1.355422). at this speed I expected the Cl to be a lot bigger as shown in table 7. The rate that the Cl and Cd increase is very comparable to the transonic velocity which could indicate that velocities closer or greater than the speed of sound are directly proportional.

Figure 11 demonstrates the velocity vectors at 10 degrees. There is large area above leading behind the wing that has a high velocity. It can also bee seen that there is a high velocity below the trailing edge which implies that there is not much drag force. Behind the trailing edge there is an section of low velocity suggesting it's a wake flow.

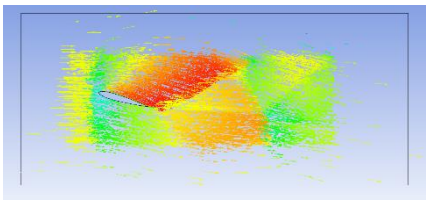


Figure 11

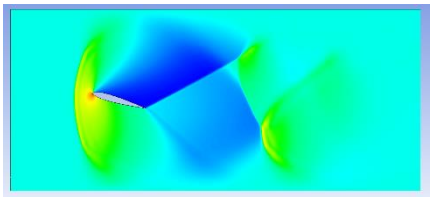


Figure 12

Figure 12 further demonstrates the wake flow behind the trailing edge as we can see that there is a low-pressure zone. We can also see that there are a few shockwaves located o the trailing edge and behind the low-pressure zones. These shockwaves are formed when supersonic flow encounters and inwards corner.

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3D Results and their interpretation – Task 4

AoA							
0 Deg		2 Deg		6 Deg		10 Deg	
Result							
X(A) - Force	152.199	X(A) - Force	140.428	X(A) - Force	34.2176	X(A) - Force	-181.145
Y(N) - Force	49.0507	Y(N) - Force	715.029	Y(N) - Force	2070.39	Y(N) - Force	3476.73
Z - Force	2.69238	Z - Force	-1.4845	Z - Force	3.41709	Z - Force	9.69871
Z - Tourqe (Piching Moment)	54.9547	Z - Tourqe (Piching Moment)	348.238	Z - Tourqe (Piching Moment)	963.832	Z - Tourqe (Piching Moment)	1631.41
Area	0.018	Area	0.018	Area	0.018	Area	0.018
Lift	49.0507	Lift	715.029	Lift	2070.39	Lift	3476.73
Drag	152.199	Drag	140.428	Drag	34.2176	Drag	-181.145
Cl	3.620412042	Cl	52.77599712	Cl	152.8146364	Cl	256.6160148
Cd	11.23374574	Cd	10.36493306	Cd	2.525587017	Cd	-13.37023813
Cm	0.005633589	Cm	0.035699033	Cm	0.098805617	Cm	0.167241254

Table 8

Table 8 shows the aerodynamics results for a 3D wing at a subsonic velocity of 50m/s. I chose to simulate 4 different angles of attack using the formulars shown in figure 1. When creating the domain, I used the same sizes as the 2D aerofoil excepts from the sides where one side was 1/3 the length of the span (2m). When initially meshing the 3D aerofoil it would produce over 35,000,000 elements. I soon discovered that the computer that I was doing the CFD on could only handle 25,000,000 elements. To fix this issue I adjusted the inner and outer element sized so that the number of elements produced would be like my 2D element size which is roughly 470,000. The parameter for the mesh is shown in table 10.
The results in table 8 shows the Cl increasing by a large amount whilst having a fluctuating Cd. This could imply that I might have made a human error in setting up the angles of attack.

Outside Domain Size		Inside Domain Size	
Sides	Size (mm)	Sides	Size (mm)
Font	2700	Font	1200
Top	2700	Top	1200
Bottom	2700	Bottom	1200
Back	9000	Back	6000
Side 1	666	Side 1	666
Side 2	0	Side 2	0

Table 9

3D Mesh	
Inflation	
First layer hieght	8.10E-06
Maximum layers	8
Maximum Angle	180
Maximum layers	23
Outer Element size	0.6
Inner Element size	0.16
Growth Rate	1.2
Nodes	111277
Elements	467409

Table 10

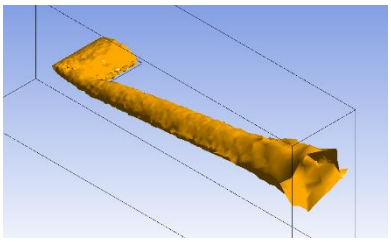


Figure 13

Figure 13 shows the vortex produced on the trailing edge at an angle of attack of 6 degrees. Vortices are a source of drag which effects the effectiveness of the aerofoil *(Refer to Reference 5)*. Figure 14 shows the vortex at 0 degrees, we can see that there is less of a vortex produced at 0 degrees this could be like this due to the aerofoil being symmetrical. There are vortices being produced along the trailing edge in figure 14 whereas in figure 13 it is just produced on the corner.

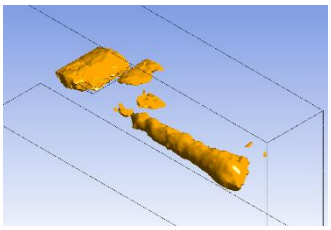


Figure 14

Proof/discussion on the accuracy of the results & what affects the accuracy

```
 ============================================================
 OUTER LOOP ITERATION =  200 (  100) CPU SECONDS = 7.013E+03 (2.522E+03)
 ------------------------------------------------------------
 |       Equation       | Rate | RMS Res | Max Res |  Linear Solution |
 +----------------------+------+---------+---------+------------------+
 | U-Mom                | 0.99 | 1.9E-04 | 1.5E-02 |       6.6E-02  OK|
 | V-Mom                | 0.98 | 5.3E-05 | 8.1E-03 |       4.4E-02  OK|
 | W-Mom                | 1.04 | 2.3E-06 | 1.7E-04 |       4.0E-01  ok|
 | P-Mass               | 0.98 | 1.4E-04 | 4.6E-03 |  4.8  4.7E-02  OK|
 +----------------------+------+---------+---------+------------------+
 | H-Energy             | 0.96 | 1.0E-04 | 2.2E-02 |  5.5  9.1E-03  OK|
 +----------------------+------+---------+---------+------------------+
 | K-TurbKE             | 0.96 | 1.3E-04 | 1.8E-02 |  5.5  9.3E-04  OK|
 | O-TurbFreq           | 0.86 | 3.2E-05 | 9.3E-03 | 10.2  1.3E-04  OK|
 +----------------------+------+---------+---------+------------------+
```

Figure 15

```
+-----------------------------------------------------------------------+
|                  Normalised Imbalance Summary                         |
+-----------------------+-----------------------+-----------------------+
|       Equation        |      Maximum Flow     |      Imbalance (%)    |
+-----------------------+-----------------------+-----------------------+
| U-Mom                 |       4.9186E+04      |        -0.0781        |
| V-Mom                 |       4.9186E+04      |        -0.0156        |
| W-Mom                 |       4.9186E+04      |         0.0012        |
| P-Mass                |       1.0930E+02      |        -0.1214        |
+-----------------------+-----------------------+-----------------------+
| H-Energy              |       9.9526E+06      |        -0.1021        |
+-----------------------+-----------------------+-----------------------+
```

Figure 16

Domain	Boundaries	
Default Domain	Boundary - Inlet	
	Type	INLET
	Location	Inlet
	Settings	
	Flow Regime	Supersonic
	Heat Transfer	Static Temperature
	Static Temperature	2.8800e+2 [K]
	Mass And Momentum	Normal Speed and Pressure
	Normal Speed	4.5000e+2 [m s^-1]
	Relative Static Pressure	0.0000e+0 [Pa]
	Turbulence	Medium Intensity and Eddy Viscosity Ratio

Figure 17

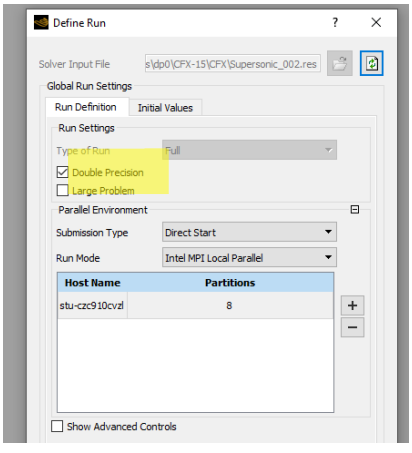


Figure 18

When starting the solution in CFX I made sure that the double precision was ticked to ensure more precise results. Once all the iterations have run, I check the Imbalance summary so see if its below 1% for a good mesh. Figure 16 shows that the supersonic at 10 degrees had a good mesh. Furthermore, the normal speed stayed the same which overall means that it was consistently accurate.

Conclusion

To conclude for the mesh study, I found that the best meshes have a larger element size as it will help to produce a more accurate set of results. Similarly, the domain sizes couldn't be too large as it would lose elements which would have an impact on the results. Then for Task 3 I found the best angle of attack at a subsonic, transonic and supersonic velocity. Which told me that the lower the angle of attack the more similar it is to the experimental data. Finally for task 4 I found that the increase in angle of attack caused a significant increase in Cl which meant the aerofoil would take off very easily. Overall, the NACA0024 is an average aerofoil which is suitable for an aerobatic plane such as the edge 540.

Graphs

Experimental Data

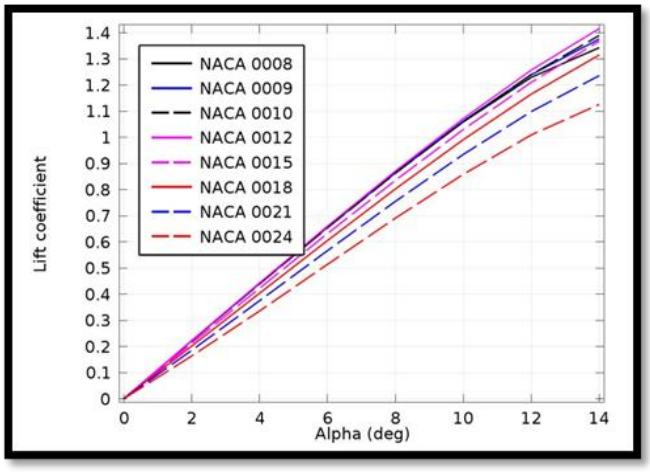


Figure 19

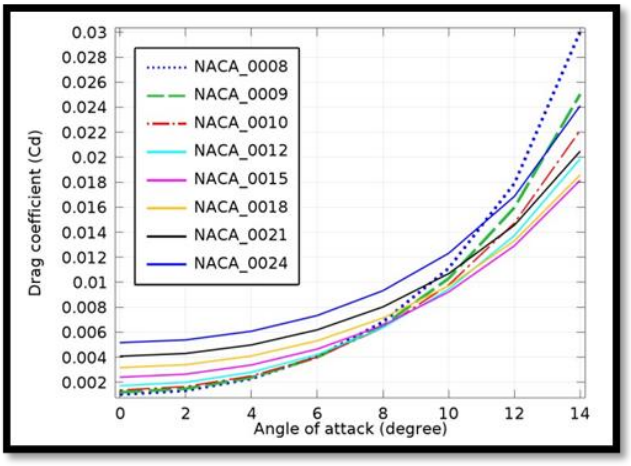


Figure 20

Figures 19 and 20 are the experimental data I was comparing my results too to see how accurate or inaccurate I was. These results were retrieved from Reference 2

Pressure Graphs

Best Mesh, Domain and Angle of Attack

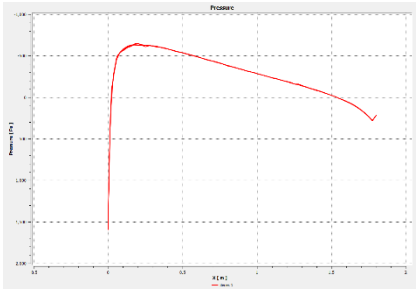


Figure 21

AoA Transonic 0 Degrees

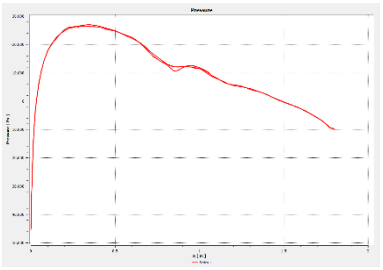


Figure 22

AoA Supersonic 0 Degrees

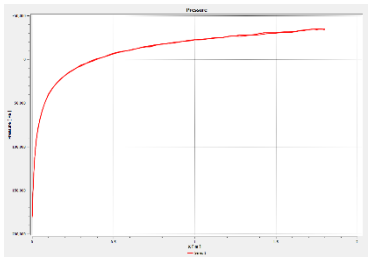


Figure 23

AoA Transonic 10 Degrees

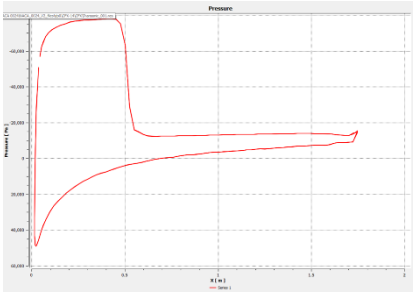


Figure 24

AoA Supersonic 10 Degrees

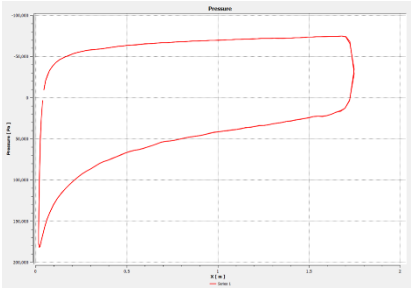


Figure 25

Figure 21-25 are all graphs extracted from the CFX results section. They represent the pressure that is put on a 2D aerofoil.

References

1. *NACA 0024 (naca0024-il) (airfoiltools.com)*
2. *SOGUKPINAR, H. (2017) "NUMERICAL SIMULATION OF 4-DIGIT INCLINED NACA 00XX AIRFOILS TO FIND OPTIMUM ANGLE OF ATTACK FOR AIRPLANE WING," Uludag University Journal of The Faculty of Engineering, 22(1).*
3. *Y+ calculator (2022) VOLUPE Software. Available at: https://volupe.se/en/wall-y-calculator-when-meshing-a-geometry-for-cfd-analysis-this-handy-volupe-calculator-computes-the-height-of-the-first-mesh-cell-off-the-wall-required-to-achieve-a-desired-y-using-flat-plate-boun/ (Accessed: January 14, 2023).*
4. Lift Coefficient - Glenn Research Center | NASA
5. Campbell-Brennan, J. (2020). *Aerodynamics: Vortices*. [online] Wavey Dynamics. Available at: https://www.waveydynamics.com/post/vortices [Accessed 15 Jan. 2023].